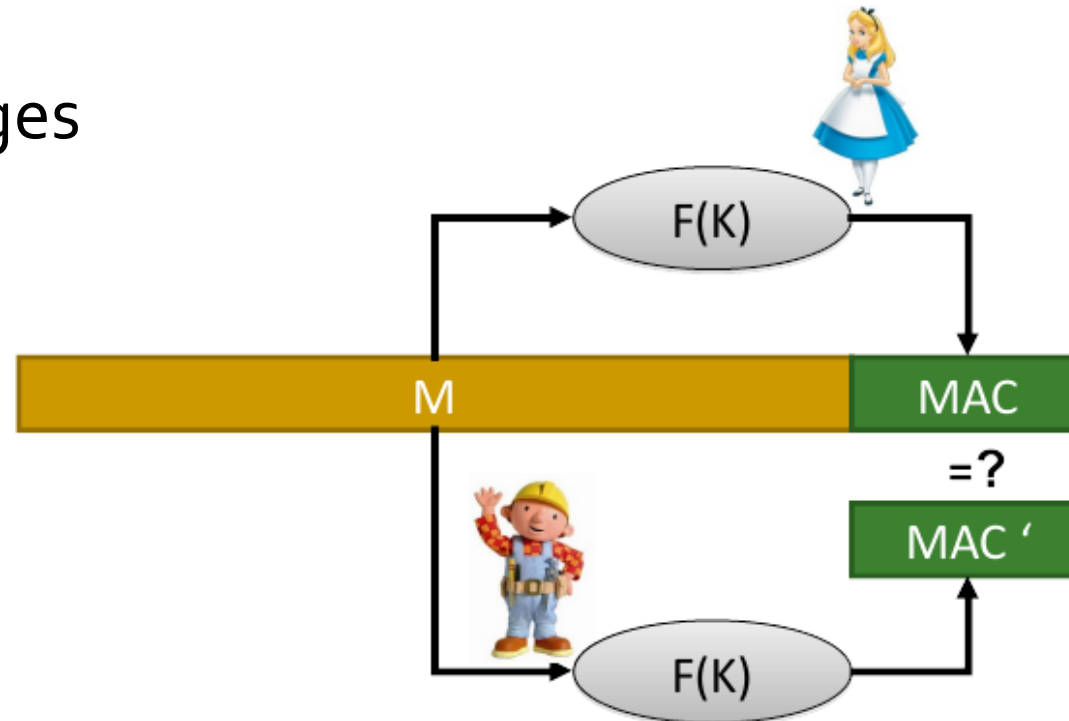




Message Authentication Codes

Message Authentication Code (MAC)

- MIC computed with a key
 - Only key holders can generate/validate the MAC
- Used to authenticate messages
 - $M' = M \parallel \text{MAC}(M)$



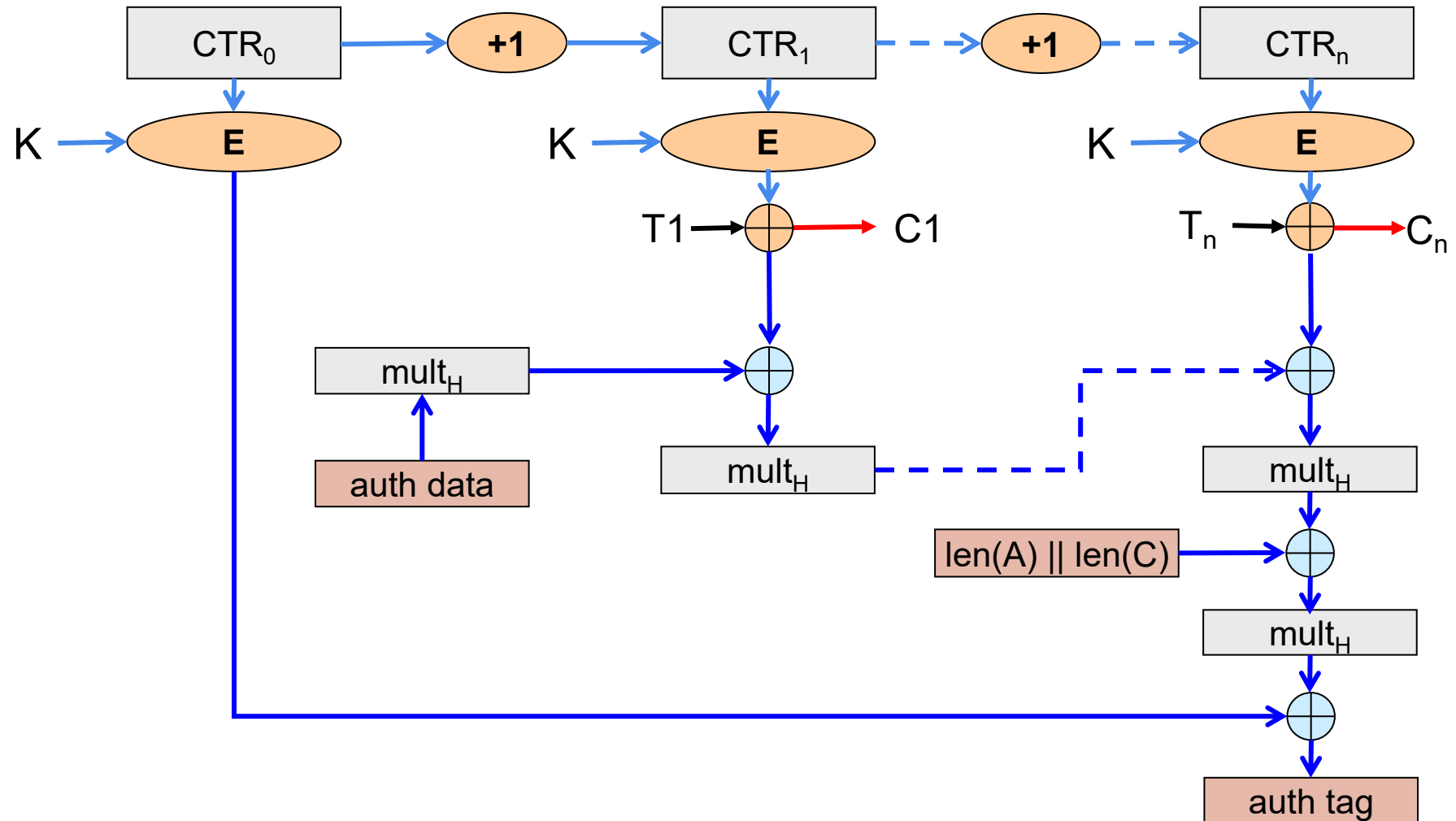
MAC: approaches

- Encryption of an ordinary digest
 - Using, for instance, a symmetric block cipher
- Using encryption with feedback & error propagation
 - CBC-MAC
- Adding a key to the hashed data
 - HMAC (output length depends on the function H used)
 - $H(K, \text{opad}, H(K, \text{ipad}, \text{text}))$
 - $\text{ipad} = \text{0x36 B times}$ $\text{opad} = \text{0x5C B times}$ $B = \text{size of H input block}$
 - HMAC-MD5, HMAC-SHA-1, etc.

Encryption + Authentication

- Encrypt-then-MAC: MAC is computed from cryptogram
 - Allows verifying integrity before (the longer) decryption
 - Preferable option
- Encrypt-and-MAC: MAC is computed from plaintext
 - MAC is not encrypted
 - May give information regarding original text (if similar to other)
- MAC-then-Encrypt: MAC is computed from plaintext
 - MAC is encrypted
 - Requires full decryption before MAC is validated

GCM (Galois Counter Mode)



Bibliografia

Leitura recomendada:

- Security in Computing, 4th edition, C. P. Pfleeger, S. L. Pfleeger: Chaps. 2
- Segurança em Redes Informáticas, A. Zúquete: Chap. 2

Leitura adicional:

- [Message authentication code](#)
- [Authenticated encryption](#)
- [Galois/Counter Mode](#)